Telco Customer Churn Analytics: Project Report

Customer retention, revenue exposure and churn drivers. Inputs: `cleaned_telco_churn_v2.csv`, `SQLQuery2.sql` *and the interactive HTML dashboard.*

1. Executive summary

Of 7,032 customers in the snapshot, **1,869 (26.6%) have churned**. Their monthly charges total **$139,131**, against **$316,530** from retained customers. Churn is concentrated, not spread evenly: month-to-month contracts, customers in their first year, fiber optic internet, customers without tech support, and electronic check payers all churn at 40% or more, while two-year contracts churn at only 2.8%.

The analysis is descriptive and based on a single snapshot. It shows where churn is high, not why it happens.

2. Business problem and objectives

A telecom provider needs to know how many customers it loses, how much monthly revenue is attached to them, and which customer groups to prioritise for retention. The objectives were to:

- Measure overall churn and the revenue associated with it.
- Identify the segments with the highest churn rates.
- Find high-value customers who have left.
- Turn the findings into practical retention actions.

3. Data overview

Item	Finding
Rows and columns	7,032 customers, 32 columns
Unique customers	`customerid` is unique, with no duplicate rows
Missing values	None

Item	Finding
Geography	All customers are in California, US
Churn definition	`churn` = 1 churned (1,869), 0 retained (5,163); identical to `churn_label`
Redundant fields	`count` is always 1; `churn_label` duplicates `churn`
Churn reason	Retained customers carry "No Churn" and must be excluded from reason analysis
Time	No reporting date, so no trends or period comparisons are possible

CLTV and churn score are used as supplied, because no documentation explains their units or how they were produced.

4. Method

1. **Inspection:** Row counts, data types, missing values, uniqueness and category distributions were checked.
2. **SQL analysis:** Ten queries in `SQLQuery2.sql` cover churn by segment, revenue and customer value.
3. **Independent validation:** Every KPI and the key SQL outputs were recalculated in Python and matched.
4. **Dashboard:** An HTML dashboard calculates all metrics live from the data, with eight filters that update every KPI, chart, insight and the customer table.

5. KPI definitions and results

KPI	Formula	Result
Churn rate	Churned customers ÷ total customers	26.6% (1,869 of 7,032)
Total customers	Distinct `customerid`	7,032 (5,163 retained, 1,869 churned)

KPI	Formula	Result
Monthly charges of churned customers	Sum of `monthly_charges` where churn = 1	$139,131
Active MRR (proxy)	Sum of `monthly_charges` where churn = 0	$316,530
Average CLTV	Mean of `cltv`	4,401 (unit not documented)
High-value churners	Churn = 1 and CLTV above the full-base average	858 customers, $64,809 per month

The $139,131 is revenue associated with churned customers in one snapshot. It is not a verified accounting loss. Active MRR is likewise a proxy, not recognised revenue. The high-value threshold is the average CLTV of the whole customer base and stays fixed when filters change.

6. Findings

6.1 Churn rate by segment

Segment	Churn rate	Customers
Month-to-month contract	42.7%	3,875
One-year contract	11.3%	1,472
Two-year contract	2.8%	1,685
Fiber optic internet	41.9%	3,096
DSL internet	19.0%	2,416
No internet service	7.4%	1,520
No tech support (internet users)	41.6%	3,472
Tech support	15.2%	2,040
Electronic check	45.3%	2,365
Mailed check	19.2%	1,604

Segment	Churn rate	Customers
Bank transfer (automatic)	16.7%	1,542
Credit card (automatic)	15.3%	1,521
Tenure 0–12 months	47.7%	
Tenure 13–24 months	28.7%	
Tenure 25–48 months	20.4%	
Tenure 49–72 months	9.5%	
Senior citizens	41.7%	1,142
Non-senior customers	23.7%	5,890
Paperless billing	33.6%	4,168
No paperless billing	16.4%	2,864
Without dependents	32.6%	5,412
With dependents	6.5%	1,620
Without online security	41.8%	3,497
With online security	14.6%	2,015

Average tenure is **18.0 months** for churned customers and **37.7 months** for retained customers.

6.2 Revenue exposure

Monthly charges of churned customers by payment method:

Payment method	Monthly charges	Churned customers
Electronic check	$84,289	1,071
Bank transfer (automatic)	$20,092	258
Credit card (automatic)	$17,947	232
Mailed check	$16,804	308

Electronic check accounts for about 61% of the $139,131. Churned customers also pay more per month on average ($74.44) than retained customers ($61.31).

6.3 Contract and internet service together

Month-to-month fiber optic customers churn at **54.6%** (2,128 customers). Two-year DSL customers churn at **1.9%** (623 customers). The combination of short contracts and fiber is the highest-risk group in the data.

6.4 Recorded churn reasons

The most frequent reasons among the 1,869 churned customers are:

Reason	Customers
Attitude of support person	192
Competitor offered higher download speeds	189
Competitor offered more data	162
Don't know	154
Competitor made better offer	140
Attitude of service provider	135
Competitor had better devices	130
Network reliability	103

Competitor-related reasons and service attitudes dominate. These are recorded reasons, not verified causes.

6.5 Churn score

The churn score averages 82.5 for churned customers and 50.1 for retained customers, so it separates the two groups. Without model documentation or performance evaluation, this shows consistency with churn, not proven predictive accuracy.

7. SQL review

All ten queries run against the actual column names and produce correct results, which were reconciled with independent calculations. Recommended wording and interpretation changes:

- **Tech support:** The column has three values. "No internet service" (7.4% churn) should be read separately from a plain "No" (41.6%).
- **High-value churners:** State the threshold (the full-base average CLTV, 4,401) in the query and results.
- **Naming:** Rename `monthly_revenue_lost` and `revenue_lost` to wording such as "monthly charges of churned customers", since they are not verified losses.
- **Casting:** The `CAST(... AS FLOAT)` calls are unnecessary but harmless.
- **Senior citizen and dependents queries:** Always report the segment size alongside the rate, as the senior-citizen query already does.

8. Dashboard summary

The dashboard is a single-page interactive report with:

- Six executive KPI cards.
- Charts for churn split, contract, tenure, revenue by payment method, high-value churners, and five churn-driver breakdowns, with the segment size shown in each tooltip.
- A churn reasons chart using churned customers as the denominator.
- Insight cards recalculated from the current filters.
- A searchable, sortable, paginated customer table with masked IDs.
- Eight filters and a reset button, with an empty-state message when no customers match.

The dashboard has no date filter or trend view, because the data has no reporting date.

9. Recommendations

1. **Move month-to-month customers to longer contracts.** Churn drops from 42.7% to 11.3% (one year) or 2.8% (two year). Test discounts or perks for annual terms, starting with fiber customers.

2. **Protect the first year.** Customers with 12 months or less churn at 47.7%. Introduce onboarding and a 90-day check-in.
3. **Review the fiber optic offer.** Fiber churns at 41.9% versus 19.0% for DSL, and competitor speed and data offers are common reasons. Review price, speed and reliability.
4. **Pilot tech support and online security bundles.** Customers without them churn at roughly 42%, versus about 15% with them.
5. **Encourage automatic payment.** Electronic check customers churn at 45.3% and hold 61% of churned monthly charges. Offer an incentive to switch.
6. **Prioritise the 858 high-value churners** for win-back and for understanding what drove them away.
7. **Review support interactions.** "Attitude of support person" is the top recorded reason.

10. Limitations

- **Single snapshot:** There are no dates, so churn cannot be tracked over time and rates are not annualised.
- **Association, not causation:** Segments overlap (for example, fiber and month-to-month), and no confounding factors were controlled for.
- **Revenue figures are proxies:** Monthly charges of churned customers are not a verified loss or a forecast.
- **Undocumented fields:** CLTV units and the churn score's method are unknown.
- **Recorded churn reasons** are not verified causes, and 154 churned customers have "Don't know".
- **Geography:** All customers are in California, so results may not generalise.
- **Dashboard testing:** The dashboard's interactions were not tested in a browser during the build, and should be checked before it is published.

11. Next steps

- Obtain dated data to measure churn trends and the effect of retention offers.
- Document the churn score and CLTV, then evaluate the score's predictive performance.
- Use a regression or similar model to separate the effects of overlapping factors.

- Run a small pilot of the contract, onboarding and payment offers, and measure the result against a control group.